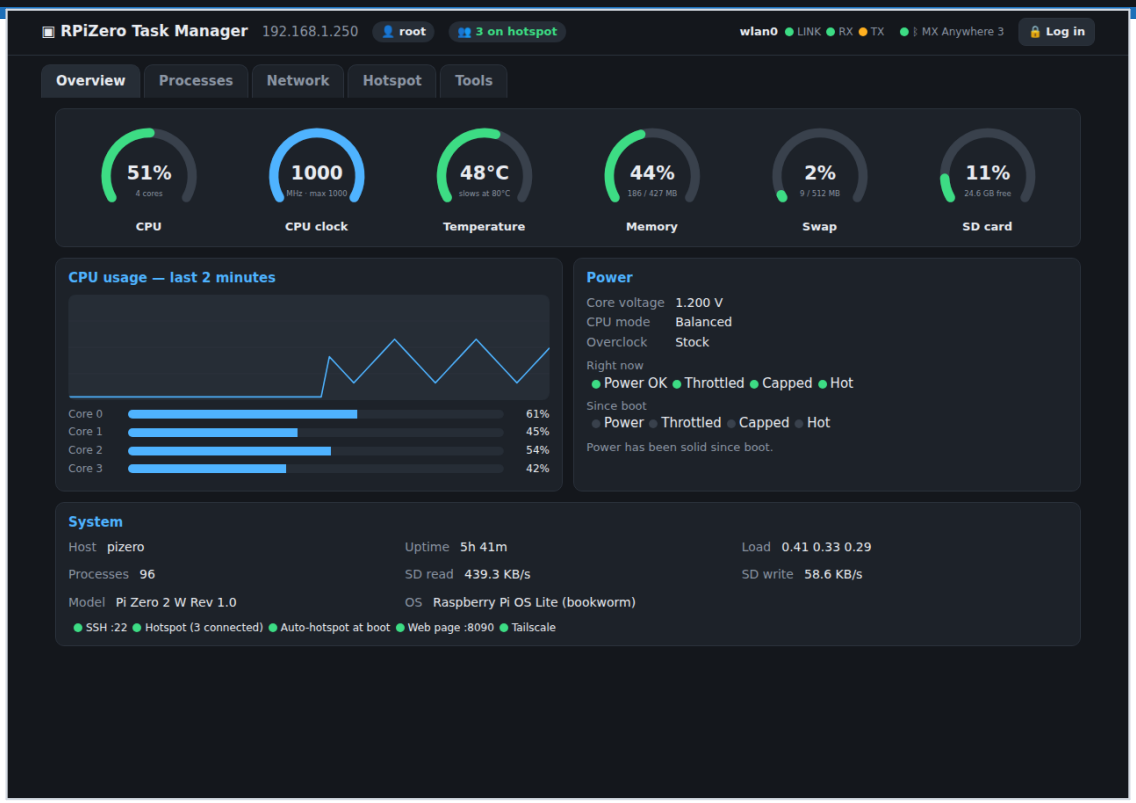
USER MANUAL · WEB EDITION

RPiZero Task Manager

A Task Manager for the Raspberry Pi Zero 2 W that you open in any web browser: stats, hotspot, visitor log, Bluetooth and tools, with no screen needed.



The Overview tab in a laptop browser

Look from anywhere	Open one web address on a phone, tablet or laptop. Nothing to install on them.
Safe by default	Anyone can see the stats; changing anything needs your control password.
Take it anywhere	Creates its own Wi-Fi, **RPIZERO-OPEN**, when there's no network, and flashes its green light while it's on
Same tools	Who's connected, visitor log, Bluetooth pairing, overclock, stability test, and Tailscale to reach it from anywhere

Version 1.0 · October 2026 · For Raspberry Pi OS Lite, Bookworm or newer

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CHAPTER 1

Welcome

The Pi Zero 2 W usually runs with no screen, so this edition of the Task Manager is a web page. The Pi serves it itself, and you open it in a browser on any phone, tablet or laptop on the same network.

Who can do what

	What they can do
Anyone on the network	See the stats: gauges, processes, network lights, and whether the hotspot is on and how many people are on it.
With the control password	Everything else: start or stop the hotspot, see who's connected and the visitor log, disconnect people, clean memory, Bluetooth, CPU mode, overclock, the stability test, end programs, and reboot or shut down.

How it differs from the desktop editions

Desktop editions	Zero edition	Why
AI tab	Not included	A useful AI model needs far more than the Zero's 512 MB of memory.
2.4 / 5 GHz switch	2.4 GHz only	The Zero 2 W's Wi-Fi has no 5 GHz.
VNC, keyboard shortcuts	This web page + SSH	Lite has no desktop. The page replaces it, and SSH is on for a terminal.
Visitor log on the Desktop	~/visitor-logs, plus downloads	There's no Desktop on Lite. Download the log or a report from the Hotspot tab.
Power and activity lights	Green light only	The Zero 2 W has a single green LED, and that's the one that flashes.

What you need

Item	Details
A Raspberry Pi Zero 2 W	With a microSD card of 8 GB or more.
Operating system	Raspberry Pi OS **Lite**, Bookworm or newer (32- or 64-bit both work).
Power	A 5 V 2.5 A micro-USB supply, or a good battery bank and cable.
Another device	Any phone, tablet or computer with a web browser, to open the page.

CHAPTER 2

Installing

You'll set up the SD card on your computer, copy the folder across, and run one command over SSH.

1. Prepare the SD card

1. On your computer, open **Raspberry Pi Imager**. Choose *Raspberry Pi Zero 2 W* and *Raspberry Pi OS Lite*.
2. Before writing, open the settings (*Edit settings*). Set a username and password, add your home Wi-Fi, and on the *Services* tab turn on **SSH**.
3. Write the card, put it in the Zero, and power it up. Give it a couple of minutes on the first start.

2. Copy the folder and install

From your Windows PC, open PowerShell in the folder holding the zip and copy it across. Use your Pi's name or address, and your username:

```
scp rpizero-taskmgr.zip hardwaremack@raspberrypi.local:~
```

Then log in over SSH and install:

```
ssh hardwaremack@raspberrypi.local
unzip -o rpizero-taskmgr.zip
cd rpizero-taskmgr
sudo bash install.sh
```

The installer asks you to choose a **control password** (6 characters or more) and type it twice. When it prints **Done!**, restart the Pi:

```
sudo reboot
```

What the installer sets up

Setup	What it means for you
The web page	Runs in the background and starts at every boot, on port **8090**.
RPIZERO-OPEN hotspot	Password **fifty-seven57**. The Pi is at **192.168.1.250**, on 2.4 GHz.
Auto-hotspot at boot	If the Pi hasn't joined a network about 60 seconds after powering on, it starts the hotspot by itself.
Flashing green light	Blinks the whole time the hotspot is on.
SSH	Turned on, so you can always get a terminal.
No password prompts	The page's controls can do their admin-level work without asking again.

Install options

Option	What it does
`WEB_PASSWORD=...`	Sets the control password without asking. If none is given and nobody's there to type one, the installer makes one up and prints it.
`WIFI_COUNTRY=US`	The Wi-Fi country, which the hotspot needs. Only used if none is set yet.
`RZ_PORT=8090`	Use a different port for the web page.
`TAILSCALE_AUTHKEY=tskey-...`	Join your Tailscale network during install, with no sign-in step (see Remote access).

Type options right after sudo, for example `sudo WEB_PASSWORD=secret123 bash install.sh`.

To update, copy the new zip across, unzip it over the old folder and run the installer again. Your password and visitor log are kept. To remove everything, run `sudo bash uninstall.sh`. Visitor logs are kept.

CHAPTER 3

Opening the page

Type one of these addresses into any browser on the same network.

Where you are	Address
On your home Wi-Fi	**http://raspberrypi.local:8090**, or the Pi's address followed by :8090 (your router's device list shows it)
On the Pi's own hotspot	**http://192.168.1.250:8090**
Anywhere else (Tailscale)	**http://100.x.y.z:8090**, the Pi's Tailscale address shown on the Tools tab. See *Remote access* in the Tools chapter.

Tip. Add the page to your phone's home screen (Share → *Add to Home Screen* on an iPhone, ⋮ → *Add to Home screen* on Android) and it opens like an app.

Logging in

1. Press **Log in** at the top right.
2. Type the control password and press **Log in**. The button changes to **Log out** and the controls appear.

- You stay logged in on that browser for 12 hours, or until you press Log out.
- After 5 wrong passwords in a row, logins pause for a few minutes. That stops anyone guessing.
- To change the password, SSH in and run `rpizero-taskmgr --set-password`.

Heads up. The hotspot password lets people onto the network, and the control password lets them change the Pi. Keep them different, and only share the hotspot one.

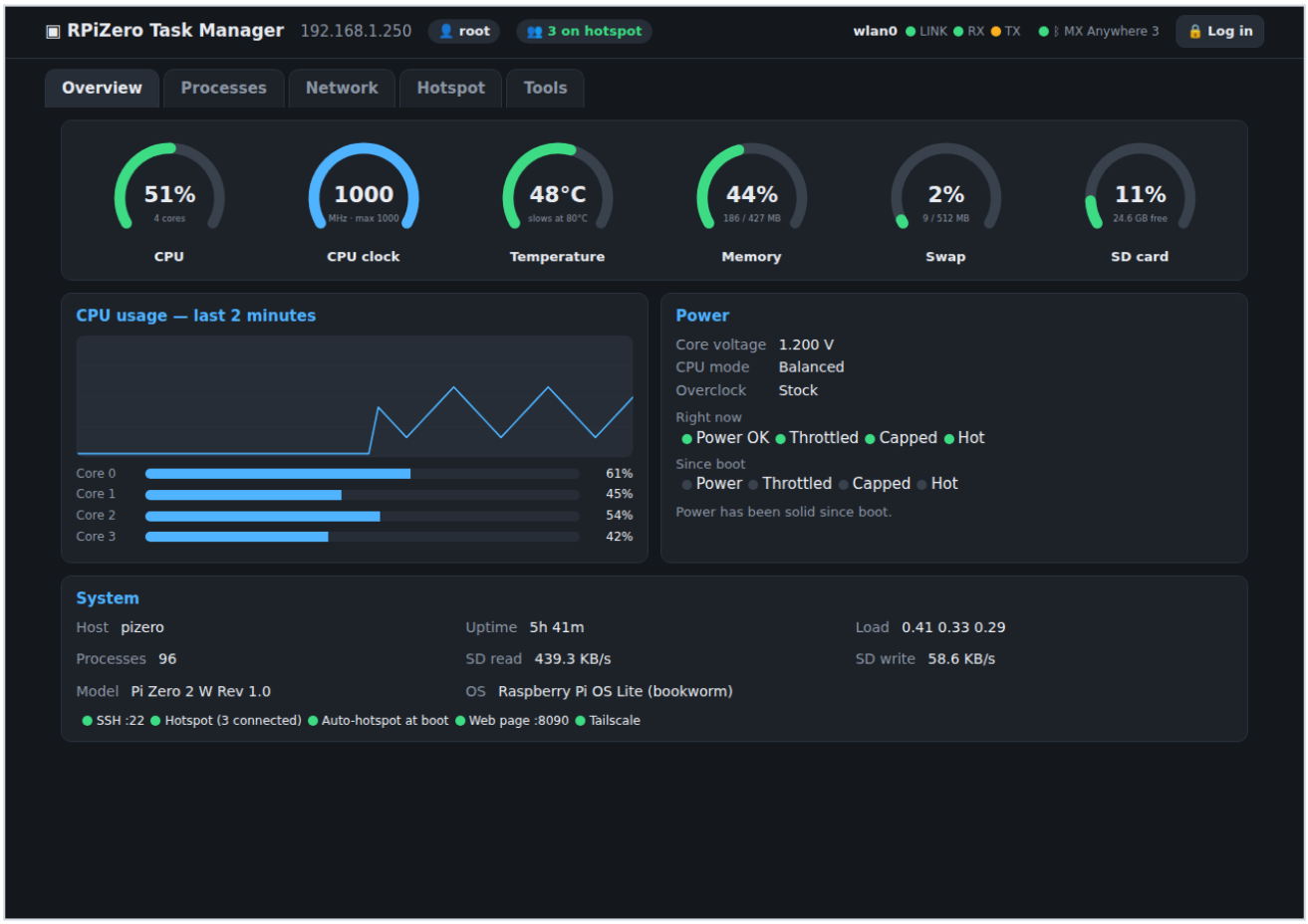
The top bar

The top bar always shows the Pi's address, the **people counter** for the hotspot (tap it to jump to the Hotspot tab), **LINK / RX / TX** lights for the Wi-Fi, and Bluetooth: green with the device name when something's connected, amber when nothing is, red when it's off. On a phone, swipe the tab row sideways to reach every tab.

CHAPTER 4

Overview tab

The Pi's health at a glance. It updates every 2 seconds.



Overview tab, as anyone on the network sees it

Gauge	What it shows	Amber / red at
CPU	How busy all 4 cores are	70% / 90%
CPU clock	Current speed (1000 MHz is the Zero 2 W's normal top speed)	no warning
Temperature	Processor temperature. It slows itself at about 80 °C	65 °C / 80 °C
Memory	RAM in use, out of about 427 MB available to programs	80% / 92%
Swap	Overflow memory on the SD card. It's slow, so keep it low	50% / 80%
SD card	How full the card is, with free space shown	80% / 92%

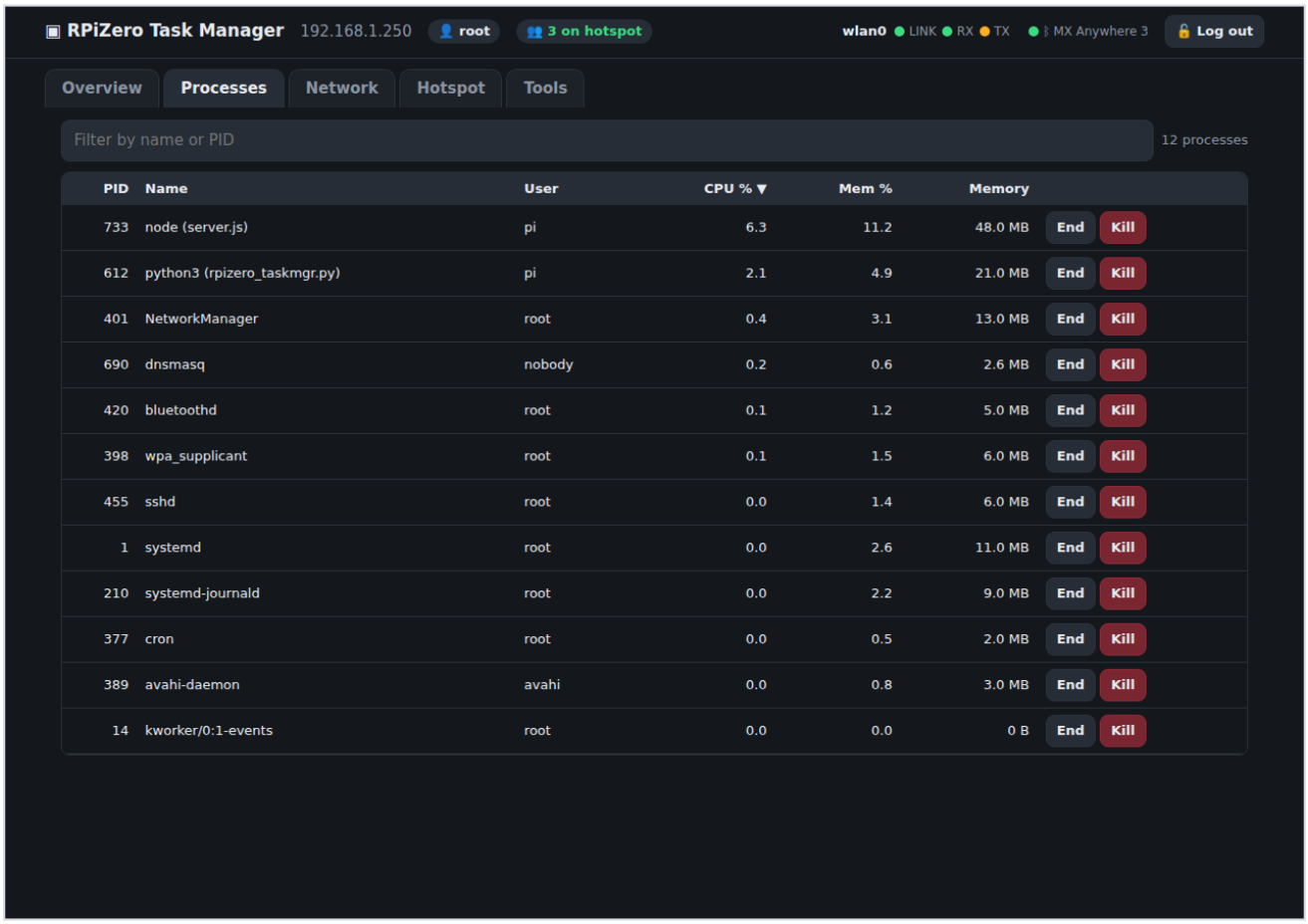
The **Power** panel shows the core voltage, CPU mode and overclock, with two rows of lights. **Right now** turns red when a problem is happening at this moment, and **Since boot** turns

amber if it happened at any time since the Pi started. The Power light is the Pi's own low-voltage alarm. The Zero can't measure its supply voltage or a battery's charge, so this light is your warning sign.

CHAPTER 5

Processes tab

Every running program, busiest first.



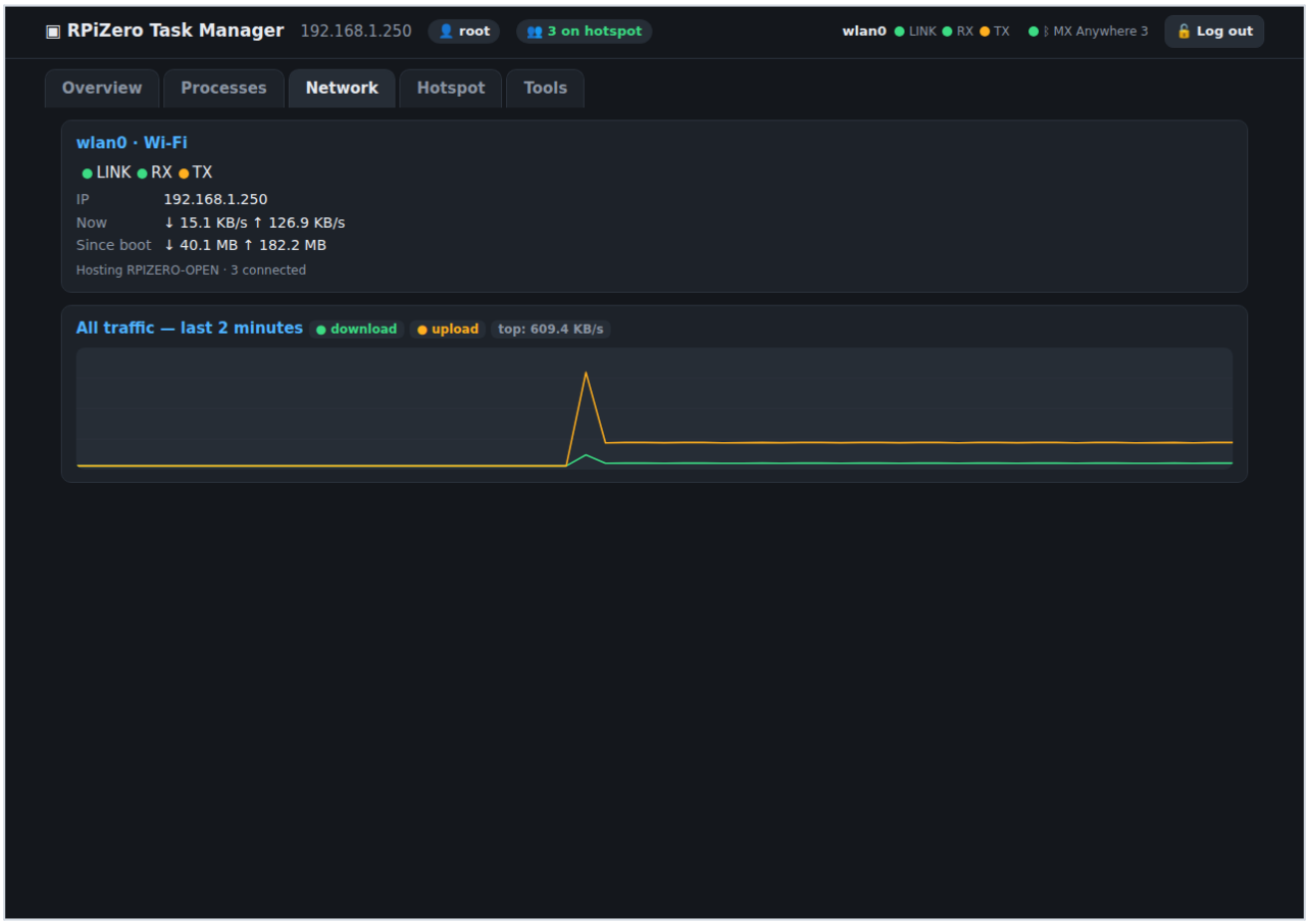
Processes tab, logged in, so End and Kill buttons show

- Tap a column heading to sort by it. Type in the filter box to find a program by name or number.
- **End** asks a program to close politely. **Kill** stops it at once, for frozen programs. Both ask you to confirm first, and both need you to be logged in.
- The list shows the 150 busiest programs, to keep the Zero light.

CHAPTER 6

Network tab

Live traffic on each connection.



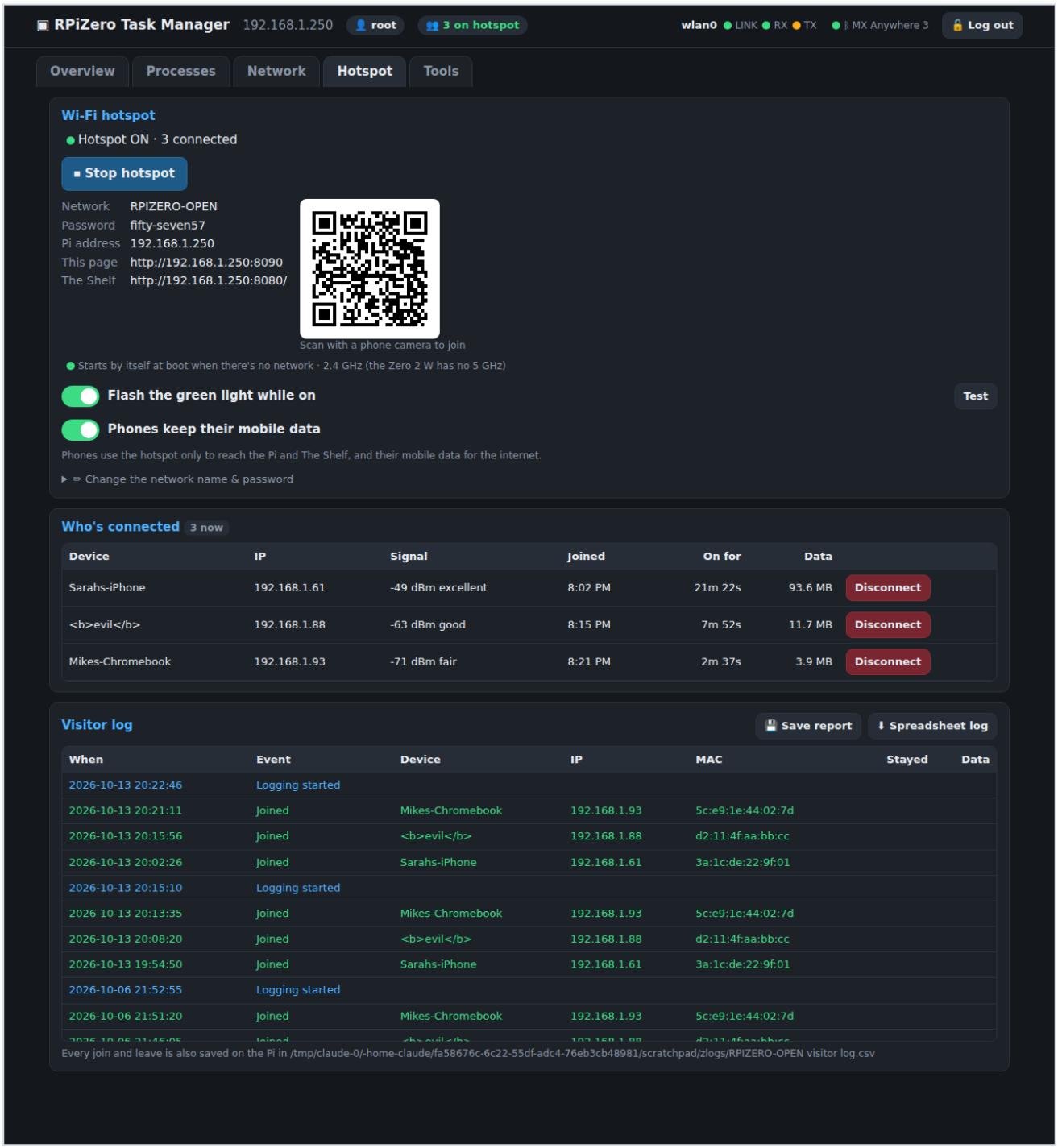
Network tab while hosting the hotspot

Each connection gets a card with blinking **LINK**, **RX** (data in) and **TX** (data out) lights, its address, current speed and totals since boot. The bottom line says which network the Pi has joined, or that it's hosting RPIZERO-OPEN. The graph shows download (green) and upload (amber) for the last 2 minutes. When people grab files from The Shelf, that's **upload** from the Pi's side.

CHAPTER 7

Hotspot tab

Run RPIZERO-OPEN, let people join, and see who's there.



Hotspot tab, logged in: hotspot controls, QR code, who's connected and the visitor log

Running the hotspot

Setting	Value
Network name	**RPIZERO-OPEN** unless you change it
Password	**fifty-seven57** unless you change it
Pi's address	**192.168.1.250**
This page on the hotspot	**http://192.168.1.250:8090**
The Shelf	**http://192.168.1.250:8080/**

- **It starts by itself:** power the Pi up away from any known network and, about 60 seconds later, RPIZERO-OPEN appears. You don't need a screen or keyboard.
- **Start or stop it** with the button on this tab. **Stop** switches it off and rejoins your saved Wi-Fi, if any is in range. It never switches the hotspot back on by mistake.
- **The green light flashes** about once a second while the hotspot is on. Turn that off with the switch, or press **Test** to see it.
- **People join** by scanning the QR code with a phone camera, or by typing the name and password.

Heads up. If you're using this page over your home Wi-Fi, starting the hotspot cuts that connection. Join RPIZERO-OPEN and open http://192.168.1.250:8090 to carry on.

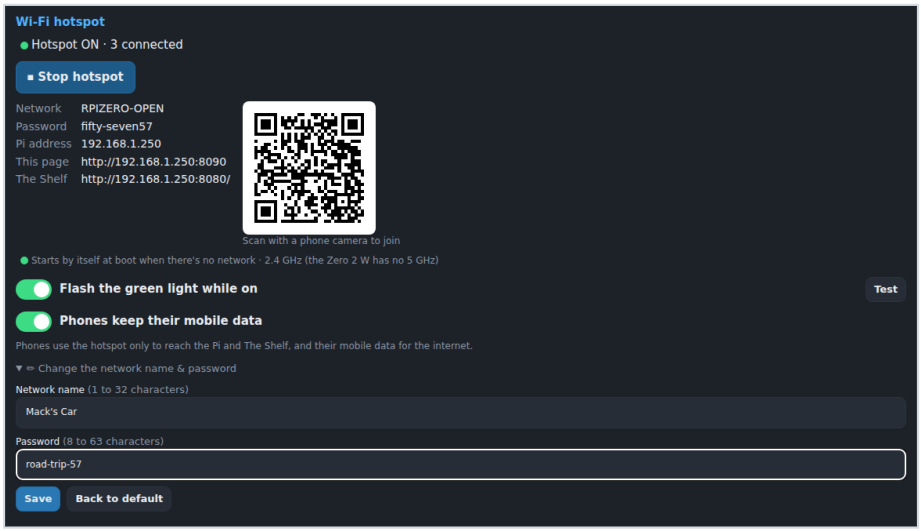
Phones keep their mobile data

The hotspot has no internet of its own. Normally a phone joined to Wi-Fi sends *everything* over it, so apps like Teams, messages and maps go quiet while you're on The Shelf. With **Phones keep their mobile data** switched on (it is from the start), the hotspot gives phones an address but no route to the internet. Each phone uses Wi-Fi only for the Pi and The Shelf, and its own mobile data for everything else.

- Switch it off only when the Zero itself is online and you want to share that connection.
- Changing it restarts the hotspot. Devices rejoin by themselves after a few seconds.
- If a phone still acts offline, forget the network on it and join again. On Android, if it asks *Stay connected?*, choose **Yes** and tick *Don't ask again*.
- Music and pages from The Shelf still come from the Pi over Wi-Fi, so they don't use your data plan.

Changing the name and password

Log in, then open ✏ **Change the network name & password** under the switches. Type a new name (1 to 32 characters) and password (8 to 63 characters) and press **Save**. **Back to default** returns to RPIZERO-OPEN and fifty-seven57.



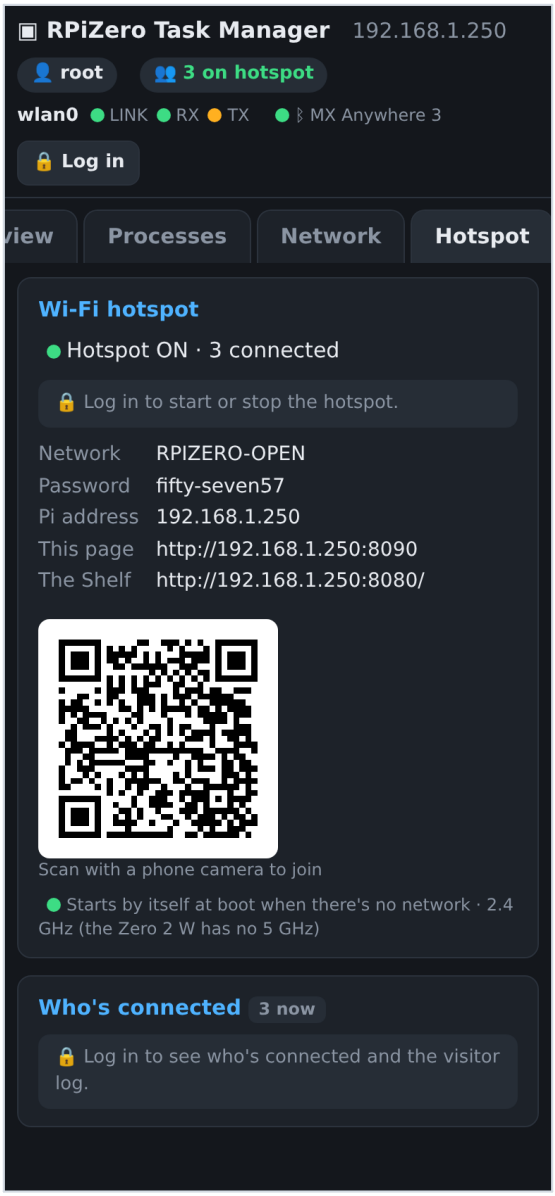
Choosing your own hotspot name and password

- The QR code updates straight away, and the automatic hotspot at boot uses the new name too.
- If the hotspot is on, it restarts with the new name, and everyone has to join again with the new password, including you if you're on this page over the hotspot.

Who's connected and the visitor log

Once logged in, you see every device on the hotspot: its name, address, signal strength, when it joined, how long it's been on and how much data it has used. **Disconnect** removes one. They can rejoin if they still have the password. Below that, the **visitor log** lists every join and leave. Joins are green, leaves are amber, and hotspot events are blue.

- Every event is also saved on the Pi in `~/visitor-logs/RPIZERO-OPEN visitor log.csv`.
- **Spreadsheet log** downloads that file to your phone or computer, and it opens in Excel or LibreOffice.
- **Save report** downloads a readable text summary: who's on now, recent activity and totals.



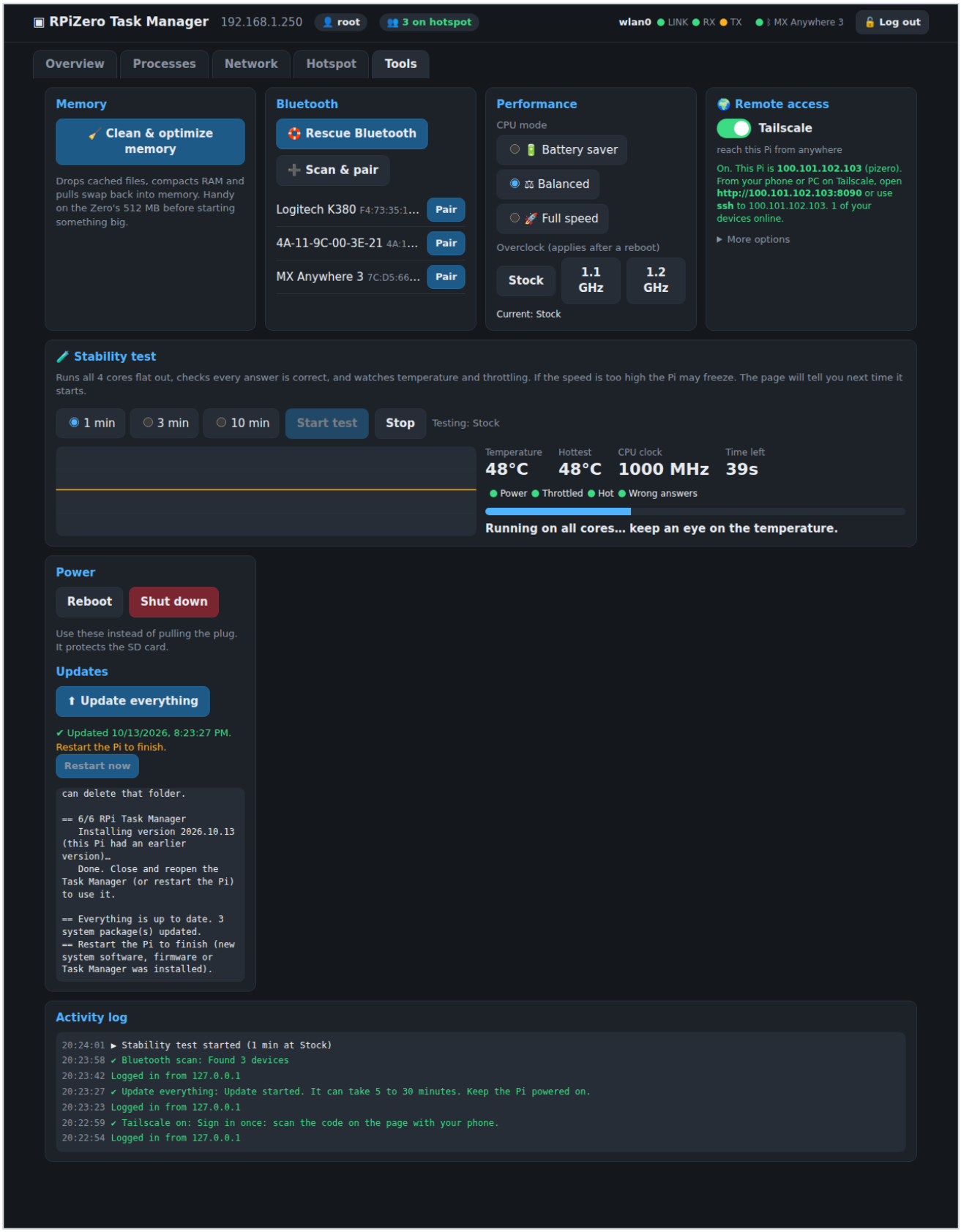
The same tab on a phone, not logged in: visitors stay private

Visitor names and addresses only show to someone who's logged in, so people on your hotspot can't see who else is there.

CHAPTER 8

Tools tab

Everything that changes the Pi. It needs you to be logged in.



Tools tab during a Bluetooth scan and a stability test

Memory

Clean & optimize memory drops cached files, compacts RAM and pulls things back out of swap. On the Zero's 512 MB it's worth doing before starting something big.

Bluetooth

- **Rescue Bluetooth** restarts Bluetooth and reconnects everything you've paired before.
- **Scan & pair** searches for 10 seconds and lists what it finds, with new devices first. Put the device in pairing mode, then press its **Pair** button. If a large code appears, type it on the new keyboard and press its Enter key.

Performance

CPU mode	What it does
Battery saver	Lowest speed: coolest, and makes a battery last longest.
Balanced	Speeds up only when needed. The normal setting.
Full speed	Always at top speed.

Overclock button	Speed	Notes
Stock	1.0 GHz	The Zero 2 W's normal speed. Use this to undo an overclock: it also turns off overclocks made by other tools or by hand.
1.1 GHz	1.1 GHz	A mild boost.
1.2 GHz	1.2 GHz	A common, well-tested boost with a small voltage bump. A stick-on heatsink helps a lot.

Overclock changes take effect after a reboot, and the page offers to reboot for you. A backup of `config.txt` is kept.

Stability test

After changing the speed and rebooting, choose 1, 3 or 10 minutes and press **Start test**. It runs all four cores flat out, checks every calculation comes out right, and shows the temperature live. You can close the page while it runs. The test keeps going on the Pi.

Result	What it means
✔ **PASSED**	No errors, no slowing down. This speed is good on your board.
⚠ **STABLE, BUT HOT**	No errors, but it slowed itself to stay cool. Add a heatsink.
⚠ **POWER PROBLEM**	The supply dipped under load. Use a stronger supply or cable.
✖ **FAILED**	Wrong answers. Choose a lower speed and reboot.

> **Heads up.** If the Pi freezes or restarts during a test, the page shows an orange warning the next time it loads, naming the speed that caused it. Step back down to the speed below it.

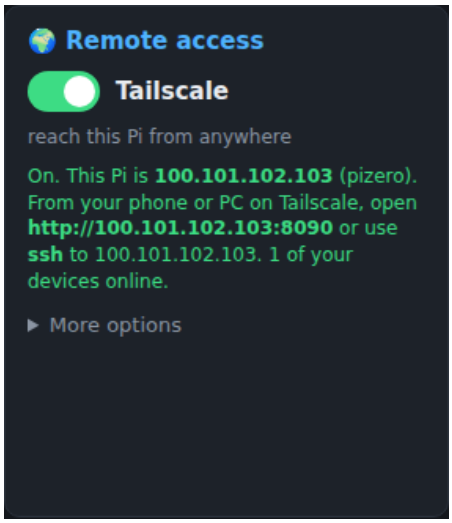
Remote access (Tailscale)

Normally the page only opens on the same network as the Pi. **Tailscale** is a free private network that links your own devices over any internet connection, with no router setup, so you can check on the Zero from mobile data or a friend's house.

1. On the Tools tab, flip **Tailscale** on. The first time it downloads itself, which needs internet and takes a minute or two on the Zero.
2. A QR code and an **Open the sign-in page** button appear. Scan the code with your phone, sign in with a Google, Microsoft, Apple or GitHub account, and press **Connect**.
3. Install the **Tailscale** app on your phone or PC and sign in with the same account.
4. The card now shows the Zero's Tailscale address, which starts with **100.** Open **http://100.x.y.z:8090** from anywhere, or use SSH to that address.



Waiting for you to sign in



Connected

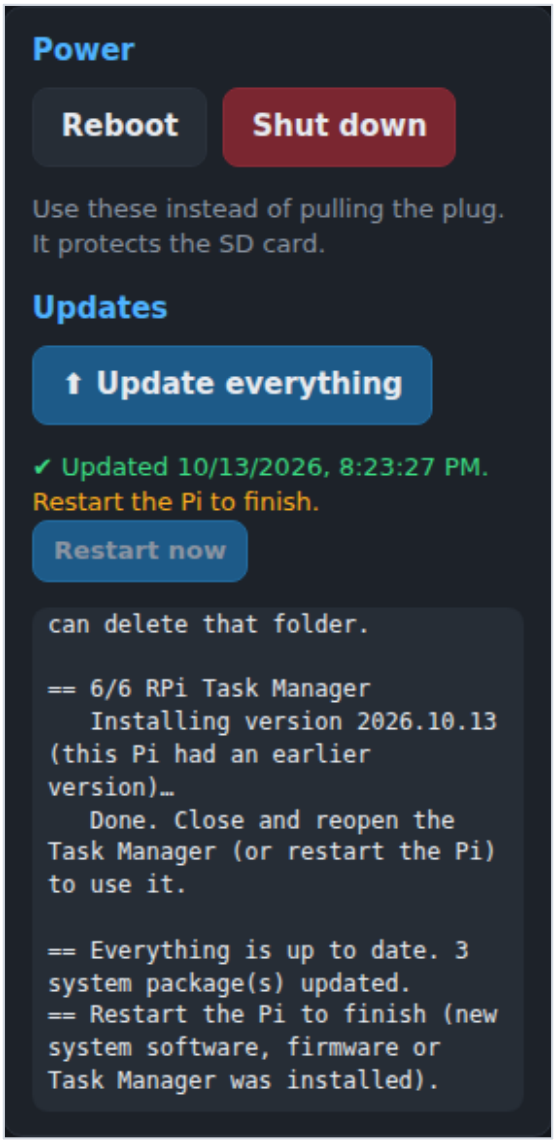
- The orange note in the first picture shows only while the Zero runs its own hotspot. It has no internet then, so Tailscale can't connect until it's back on your normal Wi-Fi.
- Switching Tailscale off keeps you signed in, so switching it back on is instant. It stays on across reboots.

- Only devices signed in to **your** Tailscale account can reach the Zero this way, and visitors who aren't logged in to the page never see the sign-in link or the address.
- Under **More options** you can join with an **auth key** from the Tailscale admin console instead, or sign this Pi out of Tailscale.

Update everything

⬆ **Update everything**, in the Power card, brings the whole Pi up to date in one go. Confirm, and the card shows each step as it happens. You can close the page; the update carries on.

Step	What it updates
1–2 System software	Raspberry Pi OS and every installed program, including Tailscale
3 Old versions	Removes old kernels, packages nothing needs any more, and downloaded installer files
4 Firmware	The Zero keeps its firmware on the SD card, so step 2 covers it
5 Your apps	**The Shelf**: its program files are replaced with the newest version from GitHub, and your pictures, music, notes and other files are kept. If The Shelf was running, it restarts with the new version.
6 RPi Task Manager	Installs a newer version of this app from GitHub if there is one, and removes any other edition of it (for example a desktop edition copied onto the Zero) so there's only one.



The Power card after an update, offering to restart the Pi

- It needs internet, so it won't run while the Zero is only on its own hotspot. On a Zero it can take 10 to 40 minutes; keep it plugged in.
- When new system software or a new Task Manager arrived, press **Restart now** to finish.

Power and activity log

Reboot and **Shut down** both ask first. Use them instead of pulling the plug, which protects the SD card. After a shutdown, unplug and replug the Pi to start it again. The **activity log** at the bottom lists what's been done recently and whether it worked.

CHAPTER 9

Power and battery tips

- Use a **5 V 2.5 A** micro-USB supply, or a battery bank with a short, decent cable. Long thin cables cause most low-voltage warnings.
- The Zero 2 W draws little power, often about 1 to 3 W, so even a small battery bank runs it for many hours. Try yours at home first.
- Watch the **Power** light on the Overview tab. Red *Right now* means the supply can't keep up.
- **Battery saver** mode on the Tools tab stretches runtime further.
- A small stick-on heatsink keeps the Zero from slowing itself in a warm bag or case, especially when overclocked.

CHAPTER 10

Troubleshooting

Problem	Try this
The page won't open	Check you're on the same network as the Pi and that the address ends in **:8090**. Over SSH, `sudo systemctl restart rpizero-taskmgr` restarts it.
I forgot the control password	SSH in and run `rpizero-taskmgr --set-password`.
“Too many wrong tries”	Wait a few minutes, then try again.
A control says the helper isn't installed	Run `sudo bash install.sh` again.
RPIZERO-OPEN didn't appear	Wait a full minute after power-on. It only starts when the Pi isn't on another network. Check the Wi-Fi country with `sudo raspi-config`.
Teams or other apps act offline while my phone is on the hotspot	Turn on **Phones keep their mobile data** on the Hotspot tab, then forget the hotspot on the phone and join again.
I forgot the hotspot password I set	SSH in and run `sudo rpizero-helper ap-set --default` to go back to RPIZERO-OPEN / fifty-seven57.
Stop hotspot didn't reconnect home	Your home Wi-Fi may be out of range. The Pi joins it by itself when it's back.
The Power light is red	Use a stronger supply or a shorter, thicker cable.
The Pi froze during a stability test	That speed is too high. Choose the one below it and reboot.
Tailscale won't switch on	The Zero needs internet for the first download and for sign-in. It can't use Tailscale while running its own hotspot.
Can't reach the page over Tailscale	Check the Tailscale app on your phone or PC is on and signed in to the same account, and that the address ends in **:8090**.
The Pi won't start after overclocking	Put the SD card in a computer, open `config.txt` and delete the lines between the `rpizero-taskmgr overclock` markers.

CHAPTER 11

Where things are kept

File or folder	What it is
`/opt/rpizero-taskmgr/`	The web server and page
`/usr/local/bin/rpizero-helper`	Does the admin-level work for the controls
`/etc/systemd/system/rpizero-taskmgr.service`	Starts the page at boot
`/usr/local/sbin/rpizero-fallback-ap`	Starts the hotspot at boot when there's no network
`~/.config/rpizero-taskmgr/web-password.json`	Your control password, scrambled, readable only by you
`/etc/rpizero-taskmgr.conf`	Hotspot name, password and mobile-data setting, and whether the light flashes
`~/visitor-logs/`	The visitor log

The same actions work over SSH too:

```
sudo rpizero-helper ap-up        # start the hotspot
sudo rpizero-helper ap-down      # stop it and rejoin your Wi-Fi
sudo rpizero-helper oc 1200      # overclock (stock|1100|1200)
sudo rpizero-helper vpn up       # Tailscale on (down | logout)
sudo rpizero-helper update start # update everything
sudo rpizero-helper ap-set 'My Car' 'new-password'   # rename (or: ap-set --default)
sudo rpizero-helper ap-data keep # phones keep mobile data (or: share)
sudo rpizero-helper led test     # flash the green light for 6 seconds
rpizero-taskmgr --set-password   # change the control password
```